

# Bitcoin Market Sentiment vs Trader Performance Analysis

## 1. Project Overview

### Project Title

Bitcoin Market Sentiment vs Trader Performance Analysis

### Objective

The objective of this project is to analyse how Bitcoin market sentiment affects trader profitability and trading behaviour.

The project combines:

- Fear & Greed Index dataset
- Historical trader data

The goal is to discover:

- How trader profitability changes with sentiment
  - Which trading strategies perform better in different market conditions
  - How trader success rate varies across Fear and Greed periods
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## 2. Business Problem

Financial markets are heavily influenced by investor emotions such as:

- Fear
- Greed
- Panic
- Optimism

This project studies whether trader performance changes based on market sentiment.

Understanding these relationships can help:

- traders improve strategies
  - firms manage risk better
  - analysts understand market psychology
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## 3. Datasets Used

### Dataset 1 — Fear & Greed Index

Columns:

- date

- classification
- value

### **Meaning**

This dataset shows the emotional state of the market.

Examples:

- Fear
  - Extreme Fear
  - Greed
  - Extreme Greed
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### **Dataset 2 — Hyper liquid Historical Trader Data**

Columns include:

- Account
- Coin
- Execution Price
- Size USD
- Side
- Closed PnL
- Timestamp IST

### **Meaning**

This dataset contains real trading activity and profitability information.

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### **4. Technologies Used**

- Python
  - Pandas
  - NumPy
  - Matplotlib
  - Seaborn
  - Jupyter Notebook
  - VS Code
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## **5. Data Preprocessing**

### **Step 1 — Load Datasets**

The datasets were loaded using pandas.

### **Step 2 — Convert Date Columns**

The timestamp columns were converted into datetime format for proper analysis.

### **Step 3 — Create Common Date Column**

A common date column was created in both datasets for merging.

### **Step 4 — Merge Datasets**

The trader dataset was merged with the sentiment dataset using the date column.

This allowed each trade to be associated with the corresponding market sentiment.

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## **6. Feature Engineering**

### **Profitability Flag**

A new column called is profit was created.

#### **Logic**

- True → profitable trade
- False → loss-making trade

This was used for win-rate analysis.

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## **7. Exploratory Data Analysis (EDA)**

### **7.1 Sentiment Distribution Analysis**

#### **Objective**

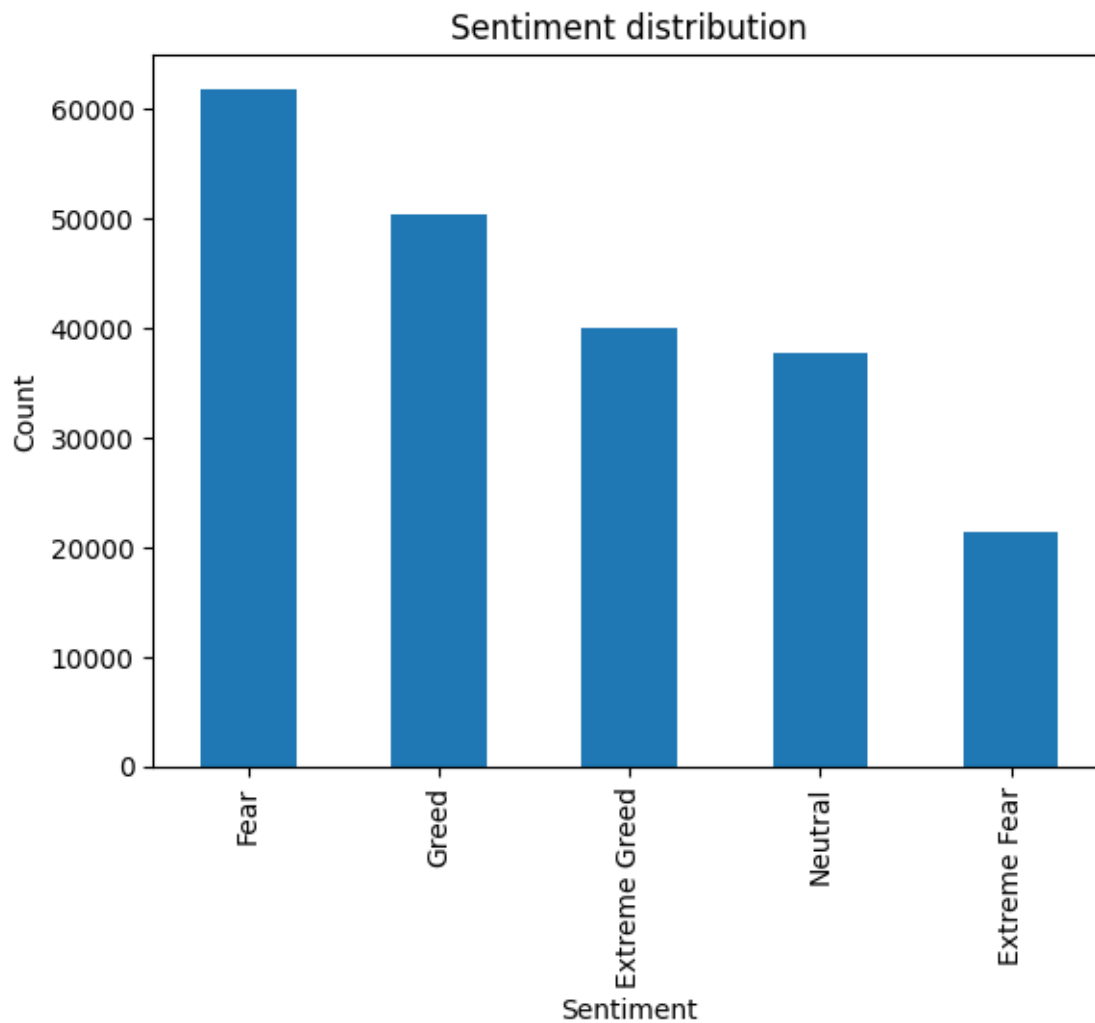
To understand how frequently different market sentiments occurred.

#### **Observation**

- Fear occurred most frequently
- Extreme Fear occurred least frequently

#### **Insight**

- Fear sentiment occurred most frequently in the dataset.
- Extreme Fear occurred the least.
- This means traders were often worried, but complete panic situations were less common.



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## 7.2 Profitability by Sentiment

### Objective

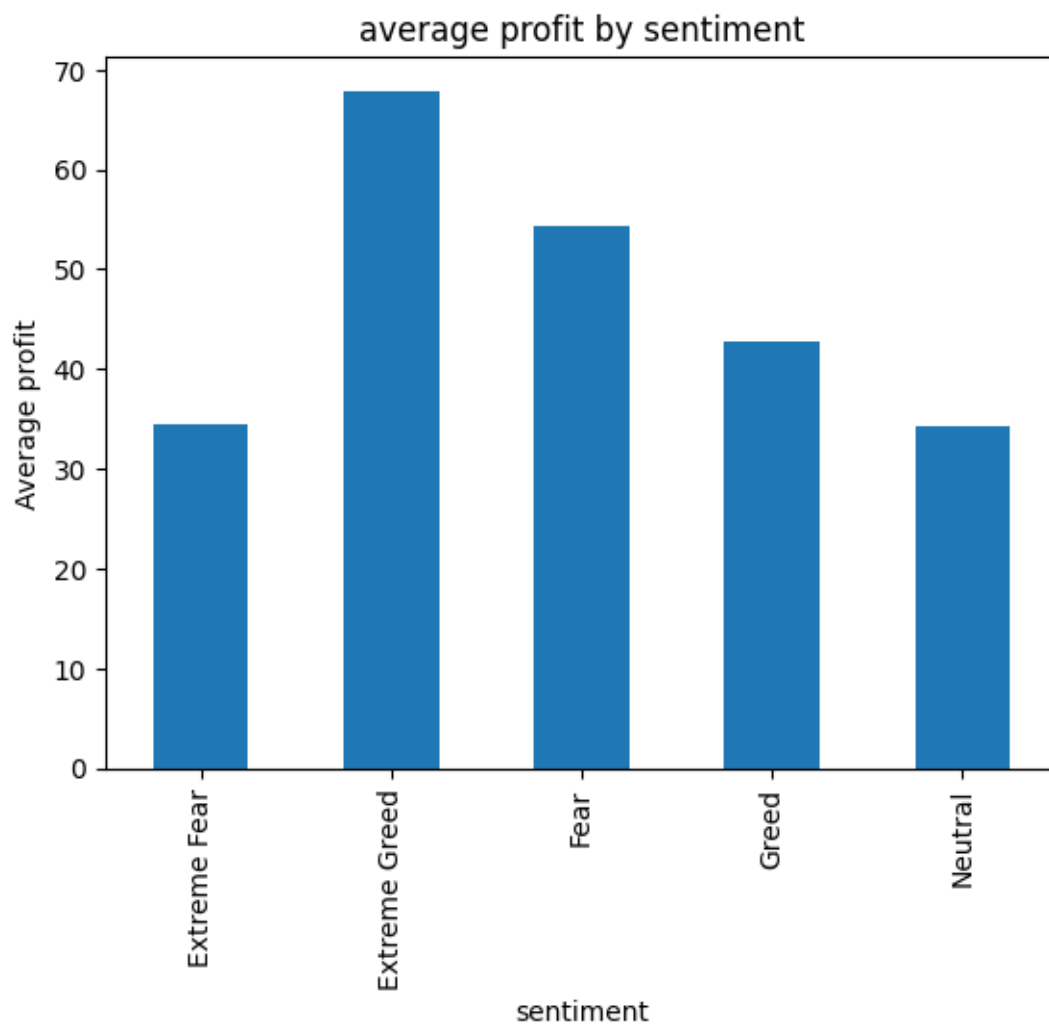
To analyze trader profitability under different market sentiments.

### Observation

- Extreme Greed showed highest average profitability
- Extreme Fear and Neutral showed lower profitability

### Insight

- Traders made the most profit when the market was in Extreme Greed.
- Lower profits were observed during Extreme Fear and Neutral markets.
- This suggests positive market sentiment improves trading performance.
- Fear markets can also create opportunities because of volatility. And because of which the experienced and confident traders make profit.



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### 7.3 Win Rate Analysis

#### Objective

To measure the percentage of successful trades under each sentiment.

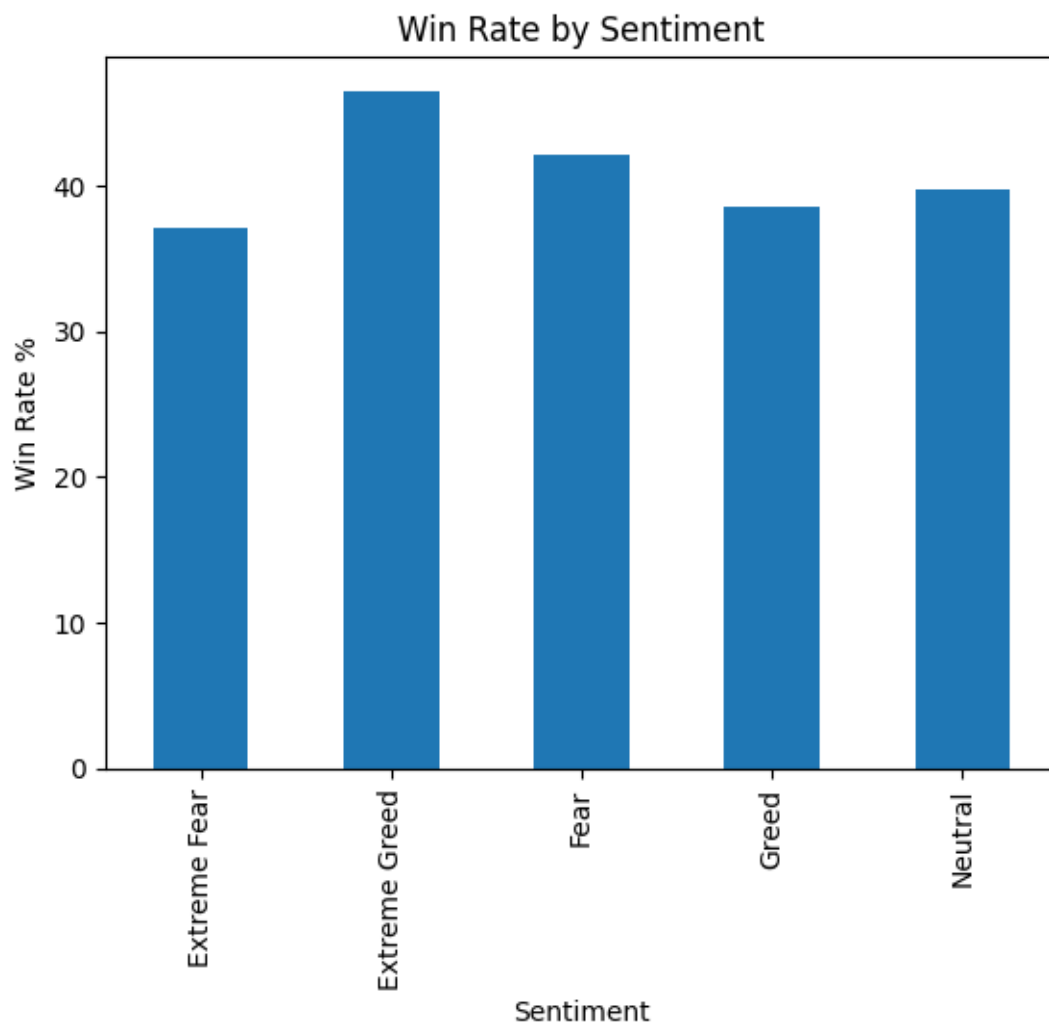
#### Observation

- Extreme Greed had highest win rate
- Extreme Fear had lowest win rate

#### Insight

- Extreme Greed markets produced the highest win rate, indicating traders perform most successfully during strongly bullish conditions.
- Extreme Fear markets showed the lowest success rate, suggesting panic-driven markets reduce trading consistency and accuracy.

- Although win rates remained below 50% overall, positive profitability indicates that successful trades may generate significantly larger returns than losing trades.



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## 7.4 Long vs Short Strategy Analysis

### Objective

To compare BUY and SELL strategy performance under different sentiments.

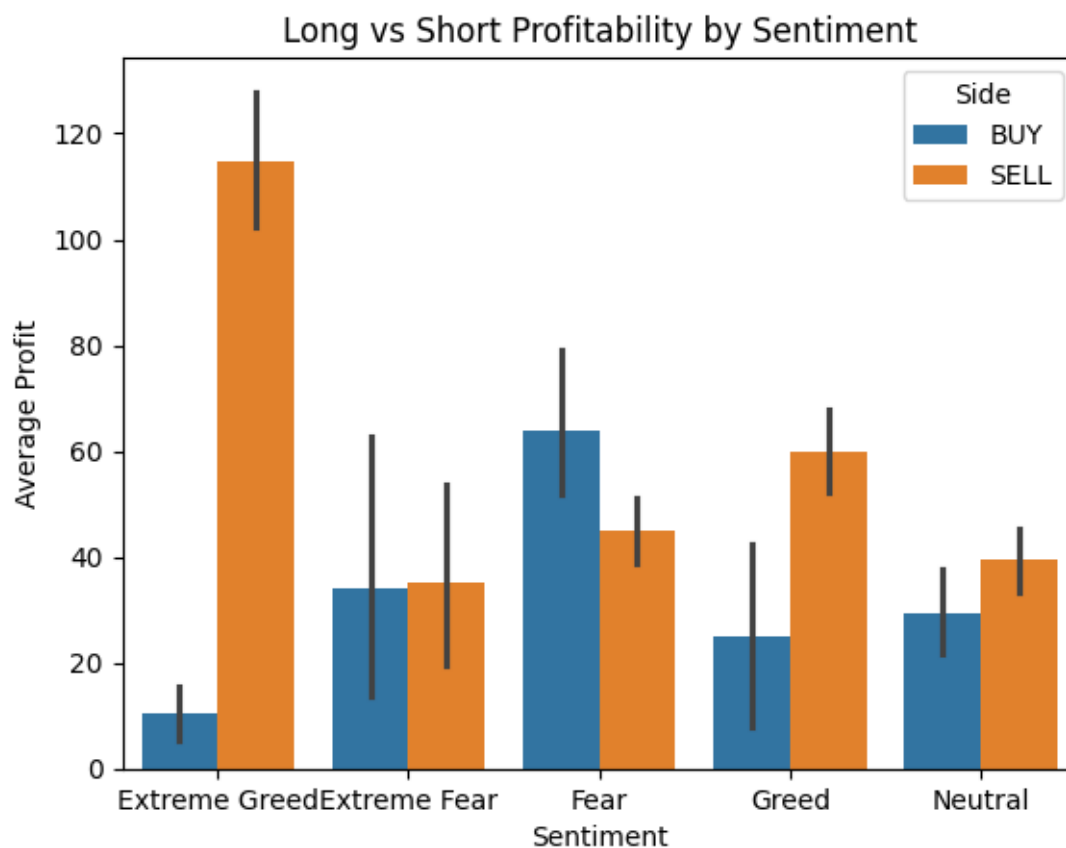
### Observation

- SELL trades performed exceptionally well during Greed and Extreme Greed
- BUY trades performed best during Fear conditions

### Insight

- Experienced traders appear to profit from market reversals during highly optimistic conditions.
- Fear markets may create buying opportunities where traders purchase undervalued assets.

- SELL trades worked better during Greed markets, while BUY trades performed better during Fear markets.
- This suggests traders use different strategies depending on market sentiment and price behaviour.



## 8. Key Findings

### Finding 1

Extreme Greed markets produced the highest trader profitability.

### Finding 2

Extreme Greed also showed the highest trade success rate.

### Finding 3

Fear conditions created profitable opportunities due to market volatility.

### Finding 4

SELL strategies performed better during Greed markets.

### Finding 5

BUY strategies performed better during Fear markets.

## **Finding 6**

Market sentiment strongly influences trader behaviour and profitability.

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## **9. Conclusion**

This project demonstrates a strong relationship between Bitcoin market sentiment and trader performance.

Key conclusions:

- Positive market sentiment improves profitability and win rate
- Fear-driven volatility creates opportunities for experienced traders
- Trading strategy effectiveness changes depending on market conditions
- Trader psychology plays a significant role in market behaviour

The analysis highlights the importance of sentiment-aware trading strategies in cryptocurrency markets.

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## **10. Future Improvements**

Possible future enhancements:

- Machine learning models for profit prediction
  - Coin-wise profitability analysis
  - Time-series forecasting
  - Risk-adjusted performance metrics
  - Dashboard creation using Power BI or Streamlit
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